

Week 3 - Physics of the Atmosphere: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Useful constants: $R_d = 287 \text{ J kg}^{-1} \text{ K}^{-1}$, $c_p = 1004 \text{ J kg}^{-1} \text{ K}^{-1}$, latent heat of vaporisation $L_v \approx 2.5 \times 10^6 \text{ J kg}^{-1}$, Wien constant $b = 2.898 \times 10^{-3} \text{ m K}$.

YOU SHOULD TRY Question 1: Using the ideal-gas equation

Air has pressure 985 hPa and temperature 288 K. Calculate its density using

$$p = \rho R_d T.$$

Comment briefly on whether the result has a sensible atmospheric magnitude.

Worked answer

Convert pressure to pascals:

$$p = 98500 \text{ Pa}.$$

Then

$$\rho = \frac{p}{R_d T} = \frac{98500}{287 \times 288} = 1.19 \text{ kg m}^{-3}.$$

A density of order 1 kg m^{-3} is entirely reasonable for near-surface air, so the magnitude provides a useful check on the calculation.

IN CLASS Question 2: A rising parcel expands and cools adiabatically

A sealed flexible dry-air parcel initially has volume $V_1 = 1.00 \text{ m}^3$ at $p_1 = 1000 \text{ hPa}$ and $T_1 = 290 \text{ K}$. It rises to a level where $p_2 = 850 \text{ hPa}$. Assume the motion is dry adiabatic and use

$$T_2 = T_1 \left(\frac{p_2}{p_1} \right)^{R_d/c_p}, \quad R_d/c_p = 0.286.$$

1. Calculate the parcel temperature T_2 .
2. Use the ideal-gas relation for a fixed amount of gas, $pV/T = \text{constant}$, to calculate V_2 .
3. Calculate the parcel density at the start and at the upper level.
4. Explain physically why the parcel both expands and cools as it rises.

Worked answer

First,

$$T_2 = 290 \left(\frac{850}{1000} \right)^{0.286} = 276.8 \text{ K}.$$

For a fixed amount of gas,

$$V_2 = V_1 \frac{p_1 T_2}{p_2 T_1} = 1.00 \frac{1000}{850} \frac{276.8}{290} = 1.123 \text{ m}^3.$$

Initially,

$$\rho_1 = \frac{100000}{287 \times 290} = 1.20 \text{ kg m}^{-3},$$

and at the upper level,

$$\rho_2 = \frac{85000}{287 \times 276.8} = 1.07 \text{ kg m}^{-3}.$$

The environmental pressure falls with height, so the flexible parcel expands until its pressure remains close to the surroundings. During adiabatic expansion the parcel does work on its environment. With negligible heat input, that work comes from internal energy, so its temperature falls.

YOU SHOULD TRY Question 3: Dalton's law in the atmosphere

Dalton's law of partial pressures states that, for an ideal-gas mixture, the total pressure is the sum of the partial pressures of the individual gases:

$$p = \sum_i p_i.$$

For an ideal-gas mixture, the partial pressure of constituent i is its mole fraction x_i multiplied by the total pressure:

$$p_i = x_i p.$$

At a location where the total pressure is 1000 hPa, suppose dry air contains 78.08% N_2 and 20.95% O_2 by mole fraction.

1. Estimate the partial pressures of N_2 and O_2 .
2. Check that the two values account for almost all of the total pressure, and explain why they do not add to exactly 1000 hPa.
3. Explain why Dalton's law is useful when water vapour is added to dry air.

Worked answer

Using $p_i = x_i p$,

$$p_{\text{N}_2} = 0.7808 \times 1000 = 780.8 \text{ hPa},$$

$$p_{\text{O}_2} = 0.2095 \times 1000 = 209.5 \text{ hPa}.$$

Together they contribute

$$780.8 + 209.5 = 990.3 \text{ hPa},$$

so about 99.0% of the total pressure. The remainder is due mainly to argon, with smaller contributions from other gases.

For moist air, Dalton's law lets us treat dry air and water vapour separately. For example,

$$p = p_d + e,$$

where e is the water-vapour partial pressure.

YOU SHOULD TRY Question 4: First-law thinking for a rising parcel

Use the sign convention

$$dU = \delta Q - p dV,$$

where $p dV$ is work done by the parcel on its surroundings.

1. What does “adiabatic” mean?
2. A parcel rises, expands and does work. If the motion is adiabatic, what happens to its internal energy and temperature?
3. Why can adiabatic motion be a useful approximation for an air parcel?

Worked answer

Adiabatic means that there is negligible heat transfer between the parcel and its environment, so $\delta Q = 0$.

For a rising parcel, $dV > 0$, so $-p dV < 0$. Therefore $dU < 0$: the parcel’s internal energy decreases. For an ideal gas, internal energy depends on temperature, so the parcel cools.

The approximation is useful when vertical motion occurs on a timescale shorter than that required for substantial conductive/radiative heat exchange with the surrounding air. It does not mean the parcel is thermally isolated forever; it is a process approximation.

EXTRA CHALLENGE Question 5: Sensible heating of air

How much energy is required to raise the temperature of 1000 kg of dry air by 5.0 K at constant pressure? Use $c_p = 1004 \text{ J kg}^{-1} \text{ K}^{-1}$.

Worked answer

At constant pressure,

$$Q = mc_p \Delta T.$$

Thus

$$Q = 1000 \times 1004 \times 5.0 = 5.02 \times 10^6 \text{ J}.$$

So about 5.0 MJ is required.

YOU SHOULD TRY Question 6: Latent heat is a large energy reservoir

Compare the energy in Question 5 with the energy released when 10 kg of water vapour condenses. Use $L_v = 2.5 \times 10^6 \text{ J kg}^{-1}$.

Worked answer

The condensation releases

$$Q = mL_v = 10 \times 2.5 \times 10^6 = 2.5 \times 10^7 \text{ J} = 25 \text{ MJ}.$$

This is about five times the 5.02 MJ required to warm 1000 kg of air by 5 K in Question 5. It illustrates why phase changes of water can strongly influence atmospheric energetics even though water is only a minority constituent.

Modelling connection.

This stability problem connects directly to dispersion modelling: the vertical temperature structure controls whether turbulent motions are suppressed or enhanced. Students who also take EART22001 will encounter the same physics in a Gaussian-plume model, but no knowledge of that unit is assumed here. The fuller treatment of vertical profiles follows in Week 4.

IN CLASS Question 7: Potential temperature and atmospheric stability

At three heights the measured pressure and temperature are:

Height (m)	p (hPa)	T (K)
0	1000	290
500	945	287
1000	890	284

Calculate potential temperature at each level using

$$\theta = T \left(\frac{1000 \text{ hPa}}{p} \right)^{0.286}.$$

Then classify the layer as statically stable, neutral or unstable. Explain what this implies qualitatively for the vertical spread of a passive pollutant plume.

Worked answer

At the surface,

$$\theta_0 = 290.0 \text{ K}.$$

At 500 m,

$$\theta_{500} = 287 \left(\frac{1000}{945} \right)^{0.286} = 291.7 \text{ K}.$$

At 1000 m,

$$\theta_{1000} = 284 \left(\frac{1000}{890} \right)^{0.286} = 293.6 \text{ K}.$$

Potential temperature increases with height, so the layer is statically stable. A vertically displaced dry parcel retains approximately its potential temperature; in this environment it tends to become negatively buoyant if displaced upward and positively buoyant if displaced downward, resisting vertical motion. A passive plume therefore tends to show reduced vertical dispersion compared with a neutral or unstable boundary layer.

EXTRA CHALLENGE Question 8: Sunlight and terrestrial radiation

Use Wien's law,

$$\lambda_{\max} T = b,$$

to calculate the peak-emission wavelength for

1. the Sun, approximated by $T = 5778 \text{ K}$;
2. the Earth's surface, approximated by $T = 288 \text{ K}$.

State the broad part of the electromagnetic spectrum in which each peak lies.

Worked answer

For the Sun,

$$\lambda_{\max} = \frac{2.898 \times 10^{-3}}{5778} = 5.02 \times 10^{-7} \text{ m} = 0.50 \mu\text{m},$$

which lies in the visible spectrum.

For the Earth,

$$\lambda_{\max} = \frac{2.898 \times 10^{-3}}{288} = 1.01 \times 10^{-5} \text{ m} = 10.1 \mu\text{m},$$

which lies in the thermal infrared. This separation is why we often speak of incoming solar “shortwave” and outgoing terrestrial “longwave” radiation.

EXTRA CHALLENGE Question 9: From the universal gas constant to R_d

The universal gas constant is $R^* = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ and the molar mass of dry air is approximately $M_d = 28.97 \times 10^{-3} \text{ kg mol}^{-1}$. Show that the specific gas constant for dry air is close to $287 \text{ J kg}^{-1} \text{ K}^{-1}$.

Worked answer

The molar form of the ideal-gas law is $pV = nR^*T$. Since $n = m/M_d$,

$$pV = \frac{m}{M_d} R^* T.$$

Dividing by V and using $\rho = m/V$ gives

$$p = \rho \left(\frac{R^*}{M_d} \right) T.$$

Therefore

$$R_d = \frac{R^*}{M_d} = \frac{8.314}{28.97 \times 10^{-3}} = 287.0 \text{ J kg}^{-1} \text{ K}^{-1}.$$